Names as Data

The field every record about a person carries, and the three jobs it is asked to do

Democracy's Data

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Every administrative record about a person carries a name, and in many it is the only description there is. The voter registration has one, and so do the campaign filing, the court docket, the jury questionnaire and the traffic stop. None was written to describe anybody, so most hold no age, no sex and no race. They hold a name, because a clerk had to write something in the box.

So the name is asked to do work, and always the same three jobs:

1. **Identify.** Pick this person out from everybody else.
2. **Link.** Two records carry the same name, so they are the same person.
3. **Infer.** The file records no sex or race, so guess them from the name.

Each is a claim about how names are spread across the country, and nobody holding a voter file can check it from the voter file. Two administrative tabulations can. How far a name carries decides whether you are matched to the right registration, flagged as somebody else on a purge list, or counted into a disparity a court must rule on.

01 The test

The first source is the Census Bureau's tabulation of the name field in the 2020 census. It appeared in April 2026 as *Frequently Occurring First Names in the 2020 Census* and its companion for last names (https://www.census.gov/topics/population/genealogy/data/2020_names.html). Together they hold 53,615 first names covering 302,031,536 people, and 156,621 surnames covering 298,898,231.

The second is the Social Security Administration's national name file: one plain text table per year of birth from 1880 to 2025, three columns wide, name and sex and count (https://www.ssa.gov/oact/babynames/limits.html).

These two are the right instruments because they count the field itself. Every other file in this book has a name column and no way to say how common a name is, which is what all three jobs depend on. Neither was made to study names. The agency writes down a name and a birth year because it cannot issue a card otherwise; the Bureau counts residents. Both tabulations are by-products, the kind of record Section III is about.

02 The data

Each row of a census table is one name with a rank and a count. The first-name table adds `pct_female`; both add six shares, `pct_white`, `pct_black`, `pct_aian`, `pct_asian`, `pct_twoplus` and `pct_hispanic`. That is the whole record: no age, no state, no way to join a first name to a surname.

Table	Names published	People covered	Share of the 2020 census	Swept into ALL OTHER NAMES
First names	53,615	302,031,536	91.1%	18,794,706
Last names	156,621	298,898,231	90.2%	36,257,637

Three things there decide what can be asked of it. **A name is listed only if about a hundred people had it**, and everyone else, 18,794,706 people by first name, goes into a final row called ALL OTHER NAMES with an empty rank. **A first name was recorded for 91.1% of those enumerated.** The other 29,417,745 are not a random draw; they sit wherever the census fell back on administrative records or a neighbour's answer. **And the counts have been noised**, by the same disclosure-avoidance machinery as the rest of that census.

Before any of it could be counted, the field had to be made. The unedited returns held about 6.0 million distinct first names, and editing removed roughly 800,000. The largest problem was not misspelling. Over ten million entries were not names at all: PERSON, CHILD, DAUGHTER, TENANT, MRS SMITH. Left alone, PERSON would have been one of the country's most common first names.

Two repairs travel into every other file you will ever join. Punctuation, numbers, spaces and suffixes were removed, so MICHAEL, MI CH AEL and MICHAEL III all count as MICHAEL. That is right for a frequency table, and it is what a matching program has to do too, which most do not. And Spanish names holding Ñ reached the Bureau with the character already turned into a question mark, so PEÑA is published as PENA. The damage happened upstream, and the repair is a second decision on top of it.

The noise was run independently into the sex table and the race table, so **49,089 of 53,615 first names carry a different total in the two.** Neither is wrong. No fact about how many Americans are named MICHAEL survives publication, because publication is what added the noise.

Rank	Name	In the sex table	In the race table	Gap
1	MICHAEL	3,476,721	3,476,715	6
2	JOHN	3,125,123	3,125,114	9
3	JAMES	2,996,421	2,996,420	1
4	DAVID	2,813,017	2,813,019	-2
5	ROBERT	2,759,339	2,759,332	7
6	WILLIAM	2,240,114	2,240,112	2
7	MARY	1,768,343	1,768,337	6
8	MARIA	1,652,966	1,652,964	2
9	DANIEL	1,607,193	1,607,193	0
10	JOSEPH	1,600,667	1,600,664	3

The Social Security file has different limits. It counts card applications by the birth year the applicant gave, and drops any name given to fewer than five babies in a year. Cards were first issued in 1936, so anyone born earlier appears only if they lived long enough to apply.

> **Before you read on, commit to a number.** The census publishes 156,621 surnames. **How many of them, the most common ones, would you need to cover half of everybody in the file?** Write it down, then do the same for first names, of which there are 53,615.

03 What it says about democracy

The first job is to identify, and a name is a distribution rather than a label.

Rank	First name	People	Last name	People
1	MICHAEL	3,476,721	SMITH	2,369,644
2	JOHN	3,125,123	JOHNSON	1,858,234
3	JAMES	2,996,421	WILLIAMS	1,561,395
4	DAVID	2,813,017	BROWN	1,386,083
5	ROBERT	2,759,339	JONES	1,383,250
6	WILLIAM	2,240,114	GARCIA	1,149,510
7	MARY	1,768,343	MILLER	1,129,186
8	MARIA	1,652,966	RODRIGUEZ	1,085,838
9	DANIEL	1,607,193	DAVIS	1,074,448
10	JOSEPH	1,600,667	MARTINEZ	1,039,848

It takes 317 first names to cover half the country, and 2,412 surnames.

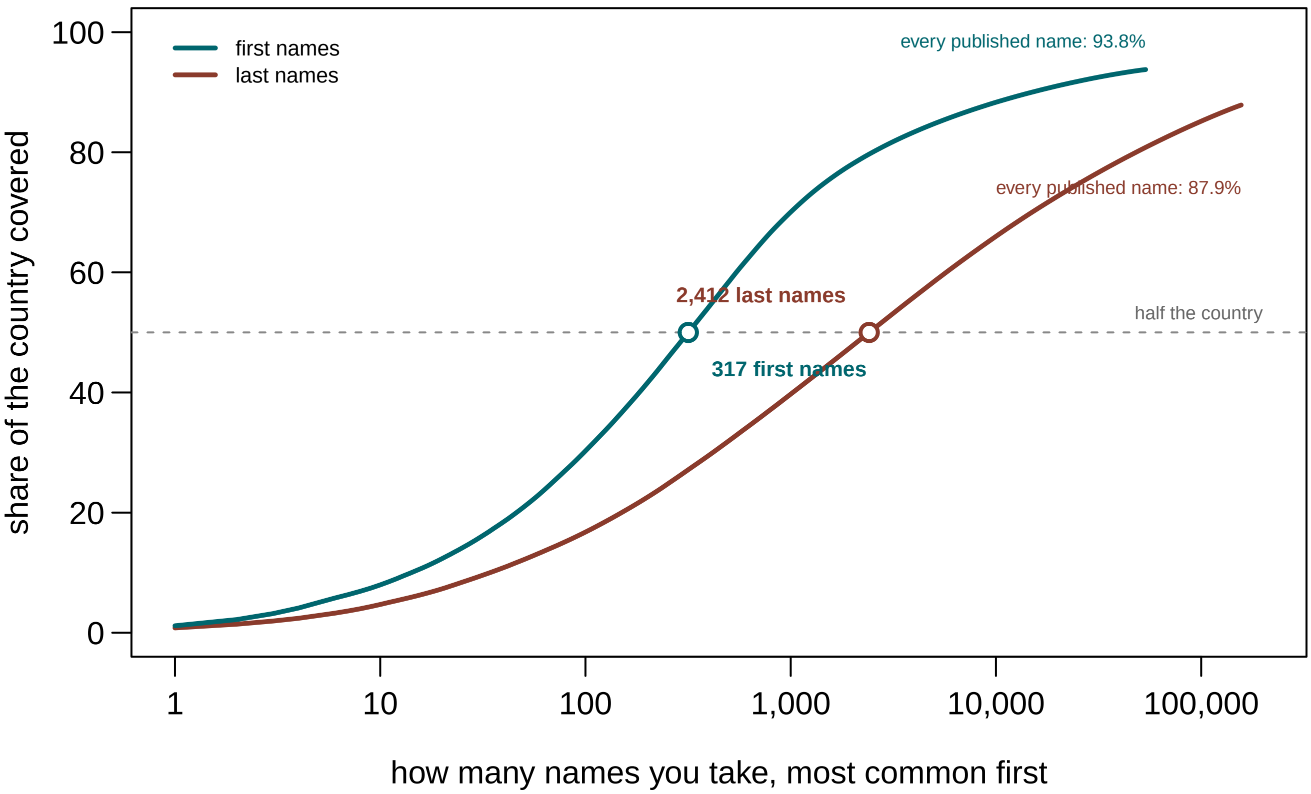

Figure 1. How much of the country the most common names cover. Rank runs along the bottom on a log scale, so the first thousand names and the hundred-thousandth are both readable. A cumulative curve fits the question, which is not what one name is worth but how fast coverage piles up. Neither curve reaches the top, and that is the threshold rather than a drawing error: every published first name covers only 93.8% of the file, every published surname 87.9%.

The most common	Of first names	Of last names
10 names	8.0%	4.7%
100 names	30.3%	16.8%
1,000 names	70.0%	39.7%
10,000 names	88.3%	66.0%

First names are the more concentrated of the two, and by a lot. A thousand first names cover 70.0% of Americans where a thousand surnames cover 39.7%. That runs against the intuition that first names are where individuality lives, and the cause is plain. A surname is inherited from a stock the country's whole immigration history filled up, and a first name is chosen from fashion, a much smaller menu.

The second job is to link, and the number underneath it is small.

Two random Americans	Chance
two people share a first name	1 in 634
two people share a last name	1 in 1,911

Two random Americans	Chance
two people share both, if the two are unrelated	1 in 1,212,277
two people share both, allowing correlation through race	1 in 1,082,518

A first name and a surname together are worth about **one in 1,212,277**. That is the figure a matching program quietly relies on, and it comes from multiplying the two separate chances. That step is legitimate only if a first name tells you nothing about a surname, and it obviously does. A floor on the error can be measured: correlate the two through the six census race categories alone, and a shared full name gets 12% more likely. The real correlation runs through national origin and generation, which is why NGUYEN and GARCIA do not draw first names from the same pot.

Each is the chance for **one pair of people**, not the chance of a match somewhere in a list, which depends on how long the list is. That is the false matches brief.

Name	As a first name	As a last name
RUSSELL	202,735	211,729
KIM	246,384	274,068
GARRETT	116,551	102,634
MORGAN	235,724	276,366
DEAN	132,116	109,762
RAY	107,593	130,289
SPENCER	106,323	135,510
GRANT	109,200	139,862
HARPER	92,376	119,191
ROSE	196,309	149,763
OLIVER	146,493	109,817
COLE	134,580	187,989

The two fields are not even distinct. 14,926 names appear in both tables, and **75.1% of Americans have a first name that is also somebody's surname**. That is why a swapped-column bug is so hard to see. Exchange `first` and `last` for part of a load, and three-quarters of the affected rows still hold an ordinary name.

The third job is to infer, and it is the one most used and least measured. Assign every person the majority sex of their first name and you are right **97.51% of the time**. That is **7,042,013 people wrong**, a population the size of Massachusetts.

Name	People	Female	Male
JESSIE	104,536	56.2%	43.8%
RILEY	191,103	56.4%	43.6%
CASEY	165,300	42.7%	57.3%
PEYTON	113,474	60.8%	39.2%

Name	People	Female	Male
DAKOTA	102,684	29.3%	70.7%
JACKIE	128,298	71.4%	28.6%
JAIME	158,232	28.5%	71.5%
AVERY	175,333	72.6%	27.4%
JORDAN	452,330	27.1%	72.9%
TAYLOR	389,216	75.5%	24.5%
ANGEL	348,575	24.2%	75.8%
JAMIE	312,010	76.7%	23.3%

Those are the most-used first names closest to an even split. JORDAN is 27.1% female and TAYLOR is 75.5%, so a majority rule errs by the same amount in opposite directions on the two. The census item is **sex**, asked as two boxes, and nothing here speaks to gender. The table is also a 2020 snapshot, so a name whose usage shifted is described by an average across the shift.

Race is the job the field is asked to do most, and the two halves of a name are not equally good at it. Put both on one scale: how much of your uncertainty about a person's race does the name remove?

Given only a	Modal guess is right	Bits before	Bits after	Uncertainty removed
First name	74.2%	1.64	1.19	27.7%
Last name	79.3%	1.70	0.94	44.6%

A surname removes 44.6% of the uncertainty; a first name removes 27.7%. That is the whole justification for surname-based inference, measured rather than assumed. It is not enough alone, because a name cannot outrun the base rate. The file's most distinctive surname for American Indians is only 3.7% American Indian, because the group is 0.7% of the file. Any method has to bring something else, usually where the person lives. That is Section IV's subject: the surnames brief builds the method from this same table.

All three jobs lean on a count of everyone alive, and such a count is slow. The census file is a stock, how many there are now; the birth file is a flow, how many arrived this year. Amazon put the Echo on sale in 2015, the best year ALEXA has ever had at rank 32. By 2025 it went to 243 girls in the whole country, down 96%.

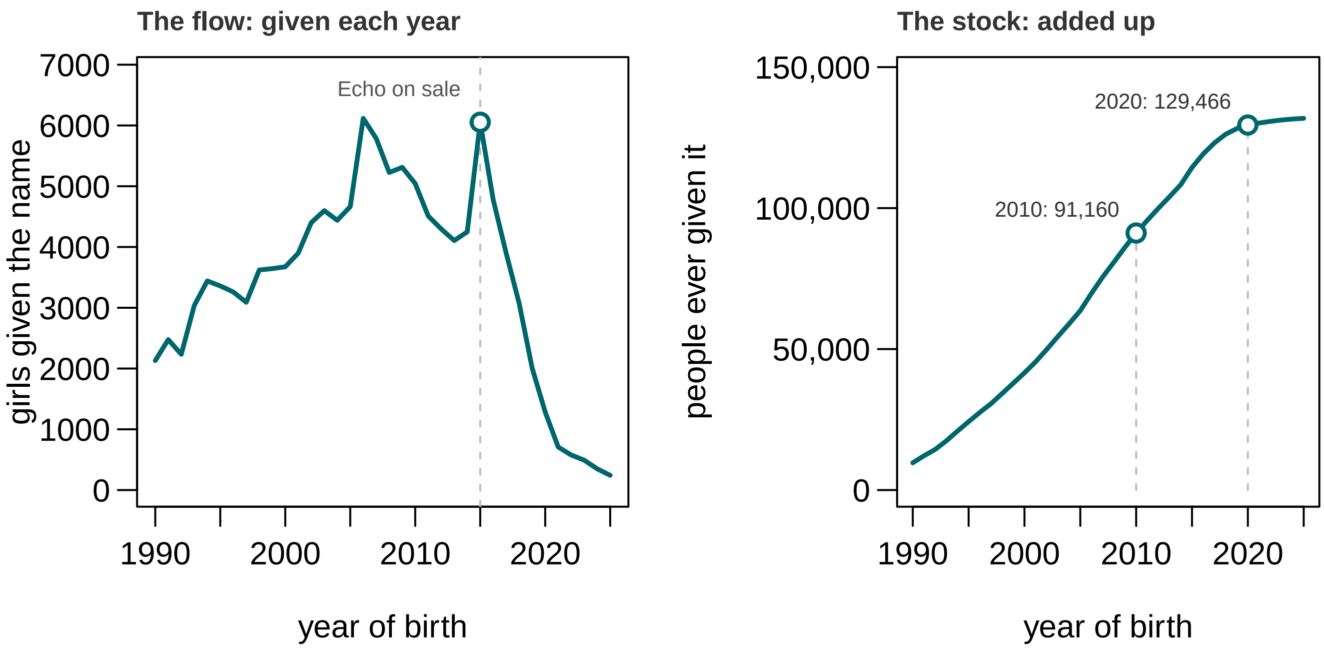

Figure 2. ALEXA, both sexes, from the Social Security birth file. The left panel is how many were given the name each year; the right is the same numbers added up, which is what a census counts. Two panels rather than two lines on one pair of axes, because a count per year and a running total share no scale.

A census taken in 2020 would have shown the name **rising by 42%**. Everyone given it before the collapse was still alive to be counted, and 38,306 more were added between the censuses. **A stock absorbs a shock and a flow records it**, and that is a rule about records rather than names. Registered voters, houses and licences are counted as stocks, and each shows a change years late and smaller than it was.

04 What this brief cannot tell you

It cannot put a first name and a surname together. They are tabulated separately, so every statement above about full names rests on multiplying two tables. The 12% correction is a floor on that error, not a measure of it.

It cannot tell you where anybody is. Names are not spread evenly and everything here is national, which is why the surname method needs a second file.

It is not a source for the racial composition of the country, and the Bureau prints that warning on both products. The shares above describe the people whose names could be used, 91.1% of the census.

It cannot see a name that arrived recently. To be published from 2020 a first name had to occur a hundred times in 2010 as well, so the names doing something interesting in the 2010s are missing by design.

And these are edited names. MICHAEL III is MICHAEL and PEÑA is PENA, which is right for a frequency table and makes the file a poor guide to the raw name field in the records you will actually join.

05 What you should have learned

- A name is a distribution, not a label: 317 first names cover half the country, and 2,412 surnames, out of 53,615 and 156,621 published.
- First names are more concentrated than surnames, 70.0% against 39.7% for the top thousand, because a first name is chosen from fashion and a surname inherited.
- A full name is worth about one in 1,212,277 for a single pair, and that is optimistic: it assumes the two halves are unrelated when they are not.
- Guessing sex from a first name is 97.51% right and 7,042,013 people wrong. A surname removes 44.6% of the uncertainty about race; a first name removes 27.7%.
- A stock absorbs a shock and a flow records it: ALEXA fell 96% in births while the census count rose 42%.

06 Extensions

None is answered above, and every one can be answered from the tables below.

- `explore_first.csv` gives every common first name a `pct_female`. Sort by it and read the names nearest 50. What happens to those people when a file guesses sex from a name?
- `explore_last.csv` has `pct_white` through `pct_hispanic` beside `people`. Sort by `pct_hispanic`, then by `people`. How many Americans do the strongest surnames cover?
- `race_info.csv` reports `bits_before`, `bits_after` and `pct_of_uncertainty_removed` for both kinds of name. Compare the rows. Is either enough to act on alone?
- `ssa_totals.csv` has `census_2020` beside `born_all` and `median_birth_year`. Pick a name once common and now rare, and say whether it is dying out or only ageing.
- **Stretch.** The Social Security file publishes a table for every state. Download two and compare one name in both, which is the geography this file cannot give you.

07 Sources

Data

- U.S. Census Bureau. *Frequently Occurring First Names in the 2020 Census*, by sex and by race and Hispanic origin, and *Frequently Occurring Last Names in the 2020 Census*. Spreadsheets, no account required. https://www.census.gov/topics/population/genealogy/data/2020_names.html
- Comenetz, Joshua. *First Name Data From the 2020 Census*, 2020 Census Briefs C2020BR-13, April 2026, and *Last Name Data From the 2020 Census*, C2020BR-14. Every statement above about editing, generic entries, the question mark and the noise is theirs. https://www2.census.gov/library/publications/decennial/2020/c2020br-13.pdf
- Social Security Administration. *Beyond the Top 1000 Names*, national data, one file per year of birth, 1880 to 2025. https://www.ssa.gov/oact/babynames/limits.html

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`both_ways.csv`, `collision.csv`, `cover.csv`, `curve.csv`, `disagreement_top.csv`, `explore_first.csv`, `explore_last.csv`, `race_extremes.csv`, `race_info.csv`, `ssa_alexa.csv`, `ssa_totals.csv`, `top25.csv`, `unisex.csv`

Related work

- Word, D. L., Coleman, C. D., Nunziata, R. and Kominski, R. (2008). "Demographic Aspects of Surnames from Census 2000." U.S. Census Bureau technical documentation, which these products descend from.
- Elliott, M. N. et al. (2009). "Using the Census Bureau's surname list to improve estimates of race/ethnicity." *Health Services and Outcomes Research Methodology* 9: 69–83, the method the surnames and BISG briefs implement by hand.

REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

```
I want to rebuild two public datasets about American names from their
original sources, without any starting files. Please do the following
and show your work.
THE FIRST SOURCE. U.S. Census Bureau, "Frequently Occurring First
Names in the 2020 Census" and "Frequently Occurring Last Names in the
2020 Census", published April 2026. Spreadsheets, served plainly from
https://www2.census.gov/topics/genealogy/2020surnames/
No account or key is needed. Fetch two of them:
Names2020_FirstNames_Sex.xlsx (about 3 MB) and
Names2020_LastNames_RaceHispanic.xlsx (about 11 MB). In each, two
header rows sit above the column names, so skip those. Read every
column as text first and convert yourself — some count columns arrive
as text, and the proportion columns carry seventeen digits. Each file
ends with one aggregate row called ALL OTHER NAMES: it is not a name,
so leave it out when counting names, but it is people, so keep it when
totaling how many people the file covers.
THE SECOND SOURCE. Social Security Administration, "Beyond the Top
1000 Names", national data: https://www.ssa.gov/oact/babynames/names.zip
(about 8 MB) — a zip holding one text file per birth year, 1880 onward,
each row a name, a sex, and a count of babies given that name.
HOW TO FETCH THE SECOND ONE. The Social Security server refuses plain
scripted downloads. Send a browser user-agent header AND accept
compressed responses — either alone still fails. If that argument
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cannot be won, download it by hand in a browser and start from the file.

WHAT TO BUILD.

1. From the census spreadsheets: how many distinct first names and how many distinct last names the Bureau published, and how many people each file covers.

2. From the Social Security files: the span of birth years, and the count of girls named ALEXA in 2015 against 2025.

CHECK YOUR WORK. The copies this book used were fetched in August 2026. If your rebuild matches them, you should find:

- 53,615 first names and 156,621 last names published
- 302,031,536 people with a first name in the census file, and 298,898,231 with a last name
- birth years 1880 through 2025 in the Social Security file
- 6,053 girls named ALEXA in 2015, and 243 in 2025

The Social Security file grows by one year every May; a later top year means the file was extended, not that you made an error.